

Convention on Prohibitions or Restrictions on the Use of Certain Conventional Weapons Which May Be Deemed to Be Excessively Injurious or to Have Indiscriminate Effects

7 April 2025

English only

Group of Governmental Experts on Emerging Technologies
in the Area of Lethal Autonomous Weapons System
Geneva, 3-7 March and 1-5 September 2025

Agenda item 5

Further consider and formulate, by consensus, a set of elements of an instrument, without prejudging its nature, and other possible measures to address emerging technologies in the area of lethal autonomous weapon systems, taking into account the example of existing Protocols within the Convention, proposals presented by High Contracting Parties and other options related to the normative and operational framework on emerging technologies in the area of lethal autonomous weapon systems, building upon the recommendations and conclusions of the Group, and bringing in expertise on legal, military, and technological aspects

Chair's summary – First 2025 session of the GGE on LAWS

Submitted by the Chairperson*

The Chairperson submits this summary, under his sole authority, with the intention of capturing his understanding of the work carried out during the first 2025 session of the Group and to maintain an informal record of proceedings to help inform the work of the Group in 2025 and ahead.

I. Background and procedural aspects

1. The first 2025 session of the Group of Governmental Experts (GGE) on emerging technologies in the area of lethal autonomous weapons systems (LAWS) (hereinafter “the Group”) was opened by the Chairperson, H.E. Mr. Robert in den Bosch of the Kingdom of the Netherlands, on Monday 3 March 2025. It was then closed on Friday 7 March 2025.
2. The Group's mandate was decided by the 2023 Meeting of the High Contracting Parties to the Convention on Certain Conventional Weapons (CCW). The Meeting decided as follows: “That the work of the open-ended Group of Governmental Experts related to emerging technologies in the area of lethal autonomous weapon systems established by Decision 1 of the Fifth Review Conference as contained in document CCW/CONF.V/10, adhering to the agreed recommendations contained in document CCW/CONF.V/2, is to continue, to strengthen the Convention. In the context of the objectives and purpose of the Convention, the Group is to further consider and formulate, by consensus, a set of elements of an instrument, without prejudging its nature, and other possible measures to address emerging technologies in the area of lethal autonomous weapon systems, taking into account the example of existing Protocols within the Convention, proposals presented by High Contracting Parties and other options related to the normative and operational framework on emerging technologies in the area of lethal autonomous weapon systems, building upon the

* The present document is being issued without formal editing.



recommendations and conclusions of the Group, and bringing in expertise on legal, military, and technological aspects.” (CCW/MSP/2023/7, paragraph 20).

3. The Group’s mandate was recalled by the 2024 Meeting of the High Contracting Parties to the CCW (CCW/MSP/2024/7, paragraph 23).

4. The Group adopted its agenda, which is contained in document CCW/GGE.1/2025/1, and considered its provisional programme of work, contained in document CCW/GGE.1/2025/2.

5. The Chairperson recalled the work which had been undertaken since the last formal session of the Group¹ (26-30 August 2024), including the in-person informal intersessional consultation for High Contracting Parties organised on 14 November 2024 to introduce the Friends of the Chair² to support him in the task of advancing the work on the rolling text. The Chairperson also briefed the Group about the virtual informal consultations for High Contracting Parties on the rolling text, held on 21 January, 27 January and 30 January, in the lead-up to the first 2025 session of the Group. The Chairperson also noted that a virtual informal consultation for observers had been held on 5 February, to exchange views with them on the rolling text.

6. The meeting heard a video message by the High Representative for Disarmament Affairs, Ms. Nakamitsu, who stated that the rapid advancement and integration of new technologies in the area of lethal autonomous weapon systems demanded States’ immediate and unwavering attention. She noted that this was also recognized in the Pact for the Future, in which Member States had decided to urgently advance the discussions on LAWS through the GGE, with the aim to develop an instrument, without prejudging its nature, and other possible measures. She stated that only a shared sense of collaboration would allow States to develop a framework that will balance their security needs with the necessary legal, humanitarian and ethical imperatives and considerations of our shared humanity

7. The Chairperson welcomed Trinidad and Tobago as a High Contracting Party to the Convention. The Chairperson also highlighted that 16 delegates were attending the first 2025 session of the Group with the support of the CCW Sponsorship Programme.

8. The session gathered 84 High Contracting Parties, 1 Signatory State, 11 Observer States, 1 organization with observer status in the United Nations, 1 United Nations entity, the International Committee of the Red Cross (ICRC), 6 academic organizations and 19 non-governmental organizations. Participants held substantive, technical and detailed discussions, the summary of which is contained in section II below.

II. Overview of discussions

9. As proposed by the Chairperson, the Group agreed to refrain from holding a general exchange of views. Rather, the Group engaged directly in substantive discussions. The Group had before it the rolling text (version as circulated on 8 November 2024, with numbers for the elements as included in the version circulated on 26 November 2024), building on the discussions held during the second 2024 session of the Group in August 2024. The rolling text is structured along 5 boxes addressing various elements, including a working characterization of LAWS, the application of IHL, prohibitions and regulations and accountability and responsibility.

A. Box I

In Box I, the Chairperson’s rolling text (version 8 November 2024) proposed the following elements:

¹ Chair’s summary – Second 2024 session of the GGE on LAWS, CCW/GGE.1/2024/WP.11.

² Claudia Henfry of Australia, Claudio Leopoldino of Brazil, Christian Hope Reyes of the Philippines, and Reto Wollenmann of Switzerland.

1. *A lethal autonomous weapon system can be characterized as an integrated combination of one or more weapons and technological components that enable the system to identify and/or select, and engage a target, without intervention by a human user in the execution of these tasks.*
2. *The above description is without prejudice to any future understanding and the potential modification of this characterization, as well as the possible exclusion of certain types of systems.*

10. The Chair stated that the working characterization remained one of the more firmly debated part of the rolling text and identified lethality, the cumulative nature (or not) of autonomous tasks, human intervention, and integration of components as key issues to be further discussed.

11. Delegations welcomed the first box of the rolling text as a meaningful step toward advancing the Group's mandate, considering it a solid foundation for further discussions and a contribution to bridging remaining divergencies between delegations. Many participants emphasized the urgency of swift action, given the rapid pace of technological advancements. To ensure the text remains relevant in the face of future technological progress, several delegations advocated for maintaining a technology-agnostic approach. In addition, many delegations argued for a broad scope of the working characterization.

12. Similar to the 2024 sessions of the Group,³ divergent opinions were expressed on the need, or not, to continue including the term “lethal” when referring to autonomous weapons systems. On the one hand, many delegations requested its deletion on the basis that “*lethality*” was an effect of a weapon and not a characterization of a weapon system, the fact that the term does not have a clear basis in IHL, the CCW and IHL also covers injuries and destruction caused to civilian object. Some of these delegations also argued that the use of “*lethal*” would narrow down the scope of the characterization too much. On the other hand, some delegates argued that the term “*lethal*” should be retained, given that it is the terminology used in the mandate of the Group. According to some of these delegations, leaving the notion of “*lethal*” out would broaden up the scope of the characterization too much. Various delegations suggested to integrate a definition of “*lethality*” into the text, which would be used in the context of the CCW and work of the Group.

13. The Chairperson proposed a definition of “*lethality*”, based on a proposal put forward by several delegations, which would be included in a new paragraph in the rolling text. Several delegations welcomed this attempt at defining the term and declared they could support it. However, others did not support the definition, as it linked “*lethality*” to the weapon itself, instead of linking it to the capacity or capability of the weapon. Many delegations also welcomed suggested text placing the definition within the scope of the application of the CCW.

14. Delegations had two rounds of detailed exchanges on the key tasks of LAWS (“*to identify and/or select, and engage*”). Some delegations were in favour of a cumulative approach, retaining only the word “*and*” between the functions (as per the language in para 19 (a) of the 2019 GGE report), whilst others deemed it essential to keep “*and/or*” in order to have a broad scope and avoid potential loopholes. This divide reflects two different visions when it comes to the elements of an instrument that the Group is working on: there are the delegations who think we need a broad characterization to ensure that all LAWS will be covered under the working characterization, whilst other delegations prefer a cumulative approach in order to avoid the risk of prohibiting/regulating weapons systems that are already used in accordance with IHL, and which are, in their view, not LAWS *per se*. Discussions also centred around the exact interpretation of the words “*identify*” and “*select*”.

³ See the Chair's summary – First 2024 session of the GGE on LAWS, CCW/GGE.1/2024/WP.6, para. 7, and the Chair's summary – Second 2024 session of the GGE on LAWS, CCW/GGE.1/2024/WP.11, para. 10.

15. Regarding the characterization of LAWS, some delegations requested to have a reference to the “activation” of the system included. According to them including “once activated” before the key tasks and “further” before “human intervention” would indicate more clearly when the autonomous process starts. Suggestions were also made to refer to “lethal force” when referring to engaging a target.

B. Box II

In Box II, the Chairperson’s rolling text (version 8 November 2024) proposed the following elements:

1. *The right of the parties to an armed conflict to choose methods and means of warfare is not unlimited.*
2. *IHL applies to armed conflict, whether international or non-international, and governs the use of all weapons, means and methods of warfare, those of the past, those of the present and those of the future, and is, consequently, applicable independent of the military technology used.*
3. *IHL, therefore, continues to apply fully to the development and use of LAWS.*
4. *In cases involving LAWS not covered by the CCW and its annexed Protocols or by other international agreements, the civilian population and the combatants at all times remain under the protection and authority of the principles of international law derived from established custom, from the principles of humanity and from the dictates of public conscience.*
5. *Context-appropriate human control and judgement with regard to the use and effects of LAWS is essential in order to ensure its employment is in compliance with international law, and in particular IHL, including the principles and requirements of distinction, proportionality and precautions in attack.*

16. Most of this section already enjoyed broad support during previous sessions of the Group. Overall, most delegations welcomed the fact that the Chairperson used language from previous reports adopted by the Group, of the Guiding Principles – notably Guiding Principle (a)⁴ - and of the preambular part of the CCW, notably the first three paragraphs.

17. Many delegations welcomed the incorporation of the Martens Clause provision, as it had been requested during the previous formal session of the Group.⁵ However, some delegations underlined that the current formulation could mean that some LAWS would fall outside of the scope of the elements discussed. In addition, some delegations recalled that the Martens Clause is already part of existing IHL and therefore proposed to delete it from the rolling text.

18. Delegations had an in-depth exchange of views on the notion of “context-appropriate human control and judgement” based on the text in section 5 of box II and on a set of questions put forward by the Chair. Some delegations reiterated their preference for using “meaningful” as a qualifier of “human control”. Most delegations reaffirmed the need for human control as a precondition for compliance with IHL and recognized the need to find a compromise solution on how to qualify human control. Other delegations repeated their concern that human control might imply direct control, which is not always required under IHL. Most delegations stated that the level of human control is depended on the specific

⁴ Guiding Principle (a): “International humanitarian law continues to apply fully to all weapons systems, including the potential development and use of lethal autonomous weapons systems.”, Annex IV to CCW/GGE.1/2019/3.

⁵ Chair’s summary – Second 2024 session of the GGE on LAWS, CCW/GGE.1/2024/WP.11, para. 17.

context and the current term “*context-appropriate human control and judgement*” includes both direct and more indirect forms of human control.

19. In a second round of interventions, following textual suggestions by the Chair, delegations engaged on the formulation of the Martens Clause. Most delegations declared that it would be beneficial to use the same language as in the preamble of the CCW, instead of formulating new language.

20. In his textual suggestions, the Chairperson refrained from amending the paragraph containing the notion of “*context-appropriate human control and judgement*” and invited delegations to further clarify their views on this concept. Some delegations reiterated that the Group should link “human” and “control” directly, but there were also clear opposing views.

C. Box III

In Box III, the Chairperson’s rolling text (version 8 November 2024) proposed the following elements:

1. *It is prohibited in all circumstances to use LAWS which are incapable of being employed in compliance with IHL, including the principles and requirements of distinction, proportionality and precautions in attack.*
2. *It is prohibited in all circumstances to employ LAWS which are of a nature to cause superfluous injury or unnecessary suffering, or which are inherently indiscriminate.*
3. *It is prohibited in all circumstances to make the civilian population as such, individual civilians or civilian objects the object of attack by LAWS.*
4. *It is prohibited in all circumstances to deploy or use LAWS if their effects in attack cannot be anticipated and controlled, as required by IHL in the circumstances of their use.*
5. *It is prohibited in all circumstances to employ LAWS that operate without context-appropriate human control and judgement.*
6. *For this purpose, parties to a conflict need to:*
 - A. *Ensure that LAWS and their effects are adequately predictable, reliable, traceable and explainable to be used in compliance with IHL.*
 - B. *Ensure that LAWS are operated with context-appropriate human control and judgment. This includes ensuring moral and legal consideration is given by a human, in particular, during identification and/or engagement stages.*
 - C. *Limit the types of targets, duration, geographical scope, and scale of the operation of LAWS, including through:*
 - i. *Ensuring that LAWS can be deactivated by a human user after activation.*
 - ii. *Ensuring that LAWS incorporate self-destruct, self-deactivation or self-neutralization mechanisms.*
 - iii. *Limiting the number of engagements that LAWS can undertake.*
 - iv. *Avoiding concentrations of civilians or civilian objects.*
 - v. *Limiting the types of targets, for instance by considering to restrict the use of LAWS against objects that are military objectives by nature.*

7. *Ensure that LAWS' mission parameters cannot be modified by the system without context-appropriate human control and judgement.*

21. Delegations continued expressing their support for an approach containing both prohibitions and regulations (the so called “two tier approach”) as a potential way forward and underlined that elements contained in box III could be core components of a possible future instrument.

22. Delegations engaged in detailed discussions on this box, including on key elements of context-appropriate human control and judgment. Regarding the paragraphs framing the “prohibitions” part, delegations were in general positive about the language used, given that it is extracted from existing IHL, and from previously agreed language of the Group. However, some delegations stated that even though using language from existing law was necessary to ensure continuity, the Group should further build on them, including through inclusion of elements specifically relevant to LAWS such as predictability, reliability, traceability and explainability. The introduction of these terms in the context of IHL led to further discussion on their precise meaning and whether they can be directly based on current IHL.

23. A number of delegations indicated that some of the regulations put forward were too prescriptive and questioned their relevance, stating that they did not consist of primary IHL obligations. However, other delegations highlighted that it was essential to have such limits in place. It was also proposed to restructure the elements listed in paragraphs 6 and 7 to better highlight the connection with existing IHL obligations. Others argued that it might be useful to move certain elements from box III to box IV.

24. In a second round of interventions, following textual suggestions by the Chair, delegations shared views on the possibility of including in the rolling text the prohibition of developing LAWS whose effects cannot be anticipated and controlled. Many delegations opposed this, stating that it could prevent the development of technology as a whole, and that it would complicate the production of such weapon systems at a too early stage.

D. Box IV

In Box IV, the Chairperson's rolling text (version 8 November 2024) proposed the following elements:

Prior to potential employment and as applicable throughout the entire life cycle of LAWS, States should:

1. *Determine, in accordance with their obligations under international law, in the study, development, acquisition, or adoption of a new weapon, means or method of warfare, whether its employment would, in some or all circumstances, be prohibited by international law. In this context, the voluntary exchange of relevant best practices between States is encouraged, bearing in mind national security considerations or commercial restrictions on proprietary information.*
2. *Ensure that LAWS are operated with context-appropriate human control and judgment across the entire life cycle of the weapon systems.*
3. *Ensure that capabilities and limitations of LAWS, in expected circumstances of use, and its anticipated effects in different scenarios are properly understood.*
4. *Ensure that weapon systems under development or modification that significantly changes the use of existing weapons systems, are reviewed to ensure compliance with IHL.*

5. *Conduct rigorous testing and evaluation in realistic simulated operational environments to enable a human user to have a reliable expectation of how the weapon system will perform in the anticipated circumstances of its use.*
6. *Conduct periodic assessments to detect and correct, as much as possible, harmful bias in data sets and implement measures to mitigate automation bias.*
7. *Implement measures to address possible harmful bias in artificial intelligence capabilities relied upon by LAWS.*
8. *States should ensure appropriate guidance, training and instructions for human users of LAWS.*

25. Many delegations welcomed the language in this box on the basis that it emphasized the importance of conducting legal reviews to ensure compliance with international law, and in particular with IHL.

26. A few delegations recalled Guiding Principle (j),⁶ and stated that it should be indicated in the rolling text that these measures should not hamper scientific and technological advancement.

27. Regarding the notion of “*context-appropriate human control and judgement*”, delegations reiterated their diverse positions shared during the exchanges on previous boxes.

28. In the previous sessions of the Group, there had been a call for stronger language on the need to prevent biases – in particular gender and racial ones – or to reduce their occurrence. Regarding this specific point, even though most delegations welcomed the inclusion of language on this risk, many recommended to use “*unintended*” instead of “*harmful*” in order to qualify it, as “*harmful*” was seen as a subjective term. One delegation also suggested to have a clearer distinction between algorithmic biases and unintended biases. Other proposed to merge the paragraphs on data and AI bias and create a separate paragraph on automation bias.

29. Several delegations requested the inclusion of an additional paragraph to address the need to prevent the proliferation and diversion of LAWS by non-States actors.

E. Box V

In Box V, the Chairperson’s rolling text (version 8 November 2024) proposed the following elements:

1. *States must ensure that humans at all times remain responsible and accountable in accordance with applicable international law for decisions with regard to LAWS.*
2. *States must ensure that human responsibility for decisions on the use of LAWS is retained since accountability cannot be transferred to machines. This should be considered across the entire life cycle of the weapon system.*
3. *States must ensure accountability across the entire life cycle of LAWS in accordance with applicable international and national law, including through the operation of such systems within a traceable and responsible chain of human command and control.*

⁶ Guiding Principle (j): “Discussions and any potential policy measures taken within the context of the CCW should not hamper progress in or access to peaceful uses of intelligent autonomous technologies.”, Annex IV to CCW/GGE.1/2019/3.

4. *States must ensure individual responsibility for the employment of means or methods of warfare involving the potential use of LAWS in accordance with their obligations under IHL.*
5. *States must ensure effective internal oversight, investigative and redressal mechanisms for the reporting and proper documentation of incidents that may involve violations of IHL. These internal mechanisms should also address incidents related to the development, deployment and operation of LAWS.*
6. *States must take all appropriate steps, including legislative and other measures, as well as imposition of penal or administrative sanctions, to prevent and suppress violations of international law by persons or on territory under its jurisdiction or control.*

30. Overall, delegations welcomed this box and there were relatively few substantive comments to the proposed text in in this box, as much of its content reflects previously agreed language. Nonetheless, technical amendments were suggested by numerous delegations, such as to change “*must*” into “*should*” throughout the box.

31. Some delegations noted that a specific reference to the notion of international responsibility of a State was absent from the text of box V, and that it would be necessary to include a paragraph on that.

32. Several delegations objected to the inclusion of the word “*traceable*” and explained that this was impractical from a technical point of view. Furthermore, some of them underlined that there is no consensus on what this term means, and that the Group should use notions which have already been discussed. Those delegations therefore requested to have only a reference to a “*responsible chain of human command and control*”.

33. Several delegations expressed their opposition to the inclusion of an obligation for High Contracting Parties to take specific legislative, penal or administrative measures regarding the restriction and/or prohibition of the use of LAWS. They suggested to delete such references on the basis that these were national prerogatives, which fell outside the mandate of the Group.

III. Way forward

34. The Group succeeded in concluding a complete reading of the rolling text, and of conducting preliminary yet in-depth discussions on all major issues. The Group also held a second round of interventions, following textual suggestions by the Chair regarding boxes 1 to 3. This first 2025 session was generally considered as productive, substantive and constructive, contributing to the delivery by the mandate of the Group.

35. At the conclusion of the session, the Chairperson informed delegations that he would share his summary of the discussions of the first session, to be followed by an updated version of the rolling text, based on the discussions during this session.

36. The Chair also announced that he would organize intersessional informal consultations before the second 2025 session of the Group in September, to facilitate moving towards a common understanding on some key notions, mindful of delegations’ capacity and the constraints of the disarmament calendar.

37. The Chairperson urged delegations to remain actively engaged and invited them to seek interaction with the Friends and the Chair’s Team during the intersessional period. The Chair concluded by reminding delegations that the Group has only three remaining sessions to finalize its mandate before the deadline of the Seventh CCW Review Conference, in 2026.